

Referee package (DRAFT) – G-A0-DUI: differentiation under the integral for the slaved variance, and A=0 uniqueness

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

This referee document concerns G-A0-DUI: the differentiation-under-the-integral justification for the slaved-variance derivative, supporting the $A = 0$ (no-condensate) uniqueness. **Scope:** the slaved-variance derivative $M'(m)$ on the window. **Not claimed:** only the DUI lemma and its consequence for $A = 0$ uniqueness.

2. The result

Lemma L1 ($M'(m) < 0$) is proven by dominated convergence with the dominating function $k^2/[m_0 + C(k^2 - q_0^2)^2]^2$, which is integrable and pointwise-dominating; G-A0-DUI is CLOSED. With L2/L3 in closed form, $A = 0$ is the unique stationary point.

3. Structure of the argument

The variance integral $M(m) = \frac{1}{2\pi^2} \int k^2/D dk$ is differentiated under the integral sign; the derivative integrand is dominated by an integrable function independent of m on the window, justifying the interchange (dominated convergence). Hence $M'(m) < 0$ pointwise, and the free-energy derivative inherits the sign that forces $A = 0$ uniqueness. The dominating lower bound $D(k, m) \geq m_0 + C(k^2 - q_0^2)^2$ holds uniformly for all m in the certified window with $m_0 > 0$, which is exactly what makes the dominating function m -independent and integrable.

4. Numerical reproduction

The sign-decomposition script reproduces $M' < 0$ and the $A = 0$ uniqueness over the window with self-test asserts; run it from the bundle. The expected output is $M'(m) < 0$ on the window and no second $A \neq 0$ stationary point.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

Result ID:	B2-PROPA-HLAYER / G-A0-DUI referee package (DRAFT)
Precise statement:	L1 ($M'(m) < 0$) is proven by dominated convergence with dominating function $k^2/[m0+C(k^2-q0^2)^2]^2$ (integrable, pointwise-dominating); G-A0-DUI CLOSED. With L2/L3 closed-form, $A=0$ is the unique stationary point.
Scope:	the slaved-variance derivative $M'(m)$ on the window; $A=0$ uniqueness.
Evidence grade:	closed-form + dominated convergence (T6/T7); EXECUTED.
Expected output:	$M'(m) < 0$ on the window; no second $A \neq 0$ stationary point found/certified.
Reproduction command:	python codes/vacuum/ha0_sign_decomposition.py
Falsifier:	a window point where $M'(m) \geq 0$; or a second $A \neq 0$ stationary point.
Tier before / after:	DRAFT referee package; DUI lemma closed. AUXILIARY DRAFT ACCEPTED (2026-06-11); not promoted to a published package.
No-overclaim statement:	the DUI lemma + $A=0$ uniqueness; not the full Prop-A comparison.
Next required action:	remain an auxiliary cited lemma (DRAFT bundle only); not promoted to a published package.